



HRITHIK RAI

Founding AI Agent Engineer | Researcher

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 [Portfolio](#)

 [Linkedin](#)

 [Github](#)

PROFILE INFO

An applied AI researcher and systems engineer with over 7 years of experience bridging advanced machine learning with production-grade engineering across government infrastructure, industrial AI labs, and international startups. My research and development portfolio spans the evolution of artificial intelligence, from early applied machine learning to large-scale agentic systems across India, Germany, and Italy. I specialize in the architecture, deployment, and optimization of highly reliable AI systems, with a core focus on compute efficiency, operational sustainability, and algorithmic rigor.

EXPERIENCE

2026 - Present

nexwise | Munich, Germany

Founding AI Agent Engineer

- Designing and deploying AI native workflows on top of complex industrial product knowledge graph

2025 - 2026

Estyl | Milan, Italy

Lead AI Engineer

- Architected a scalable, robust, highly available and responsive agentic RAG pipeline serving 500 to 2000 req/min (p95 latency = 2.8 seconds)
- End to end system design (session management via Redis, event tracking, hybrid RAG (BM25+Vector), APIs, Microservices, Data pipeline management and Tools)

2024 - 2025

Sagar Institute of Science Technology and Research | India

Assistant Professor

- Designed and taught graduate-level courses on Machine Learning, Deep Learning, and GenAI.
- Mentored and guided projects for over 300 students.

2023 - 2024

Livello Technologies | Düsseldorf, Germany

Lead Data Scientist & Researcher

- Architected a revenue and product demand forecasting system, p90 latency = 30-120 ms (single horizon, single series)
- Managed production releases on GCP, Cloud Run instances using CI/CD pipelines (Gitlab).
- Developed ETL pipelines (mongoDB, BigQuery), ensuring high quality data, integrity and accessibility for ML tasks.

EDUCATION

2021 - 2024 | Deggendorf Institute of Technology, Germany

M.S. in Artificial Intelligence and Data Science

2015 - 2019 | UIT-RGPV, India

Bachelor of Engineering, IT

SKILLS

- Python
- Agentic AI
- Generative AI
- Machine Learning
- System Design
- DevOps/MLOps

PUBLICATIONS

- [AI Driven Circular Supply Chain](#) (International Conference on Marketing 5.0 & Regenerative Economics 2025)
- [A comparison of time series models for revenue and product demand forecasting in smart fridges.](#)

2022-2023

Syskron GmbH (Krones) | Regensburg, Germany

AI Intern

- Developed self-optimizing pipelines for factory telemetry using feedback-driven learning loops, reducing drift impact by 28%.
- Co-led prototype to production transition of AI tools, managing testing, deployment, and support phases.
- Assisted factory set point setup via reinforcement learning (ContilooAI)

2018-2021

State Government of MP, India | India

Data Scientist and Analyst (Contract)

- Applied statistical analysis and SVM models to optimize resource allocation, improving system performance by 60%.
- Built chatbots using IBM Watson for policy-driven citizen services.
- Worked on structured data transformation using Python and JSON, enabling large-scale analytics.
- Built dashboards and visualizations to communicate data-driven insights to policymakers and non-technical stakeholders.

Projects

Adaptive Knowledge Engine

Reads authoritative documents into a cited, bi-temporal knowledge graph and answers three precise shapes of question against it: exact lookup, page retrieval, and configuration validity. Every answer carries a citation to its exact source page. When the engine cannot ground an answer, it stops rather than guess.

FraudHawk – Built autoencoder + Isolation Forest agent with real-time drift detection and retraining triggers.

- Evaluated feature impact using statistical methods and live KPI monitoring.

Eva: AI-Powered Health Companion – Built a real-time AI agent for evidence-based diagnostics using medical PDFs, X-rays, and clinical videos.

- Integrated LangChain + Agno Agents for retrieval-augmented generation (RAG), explainability (XAI), and on-device interaction.
- Backend: Python, Flask, ChromaDB; 300+ MAUs within 2 months.
- Demonstrated how model transparency, ethics, and performance can co-exist in medical AI.

MediSense-AI – Intelligent Personal Medicine Assistant

It is a personal medicine assistant that uses semantic search over symptom inputs to recommend medications with details on diseases, ingredients, dosage, and cost. Built with Streamlit and Pinecone, it offers fast, intelligent, and accessible healthcare guidance.

End-to-end revenue and product demand forecasting system

This SARIMAX based forecasting service predicts daily revenue and product demand for kiosks, aiding inventory management and resource allocation. It provides key insights like net sales, demand forecasts, revenue comparisons, and top-selling products/kiosks. The pipeline handles data migration, modeling, and forecasting

REFERENCES

- David Kluebers - Founder and CEO - nexwise.ai (david@nexwise.ai)
- Markus Zölfl - Innovationsmanager AI (M.Zoelfl@syskron.com)
- Victor Mita - Founder - Estyl (v.mita@estyl.ai)
- Dr. Manish Billore - Principal - Sagar Institute of Science and Technology (principal.sistec@sistec.ac.in)
- Alexander Eissing - CEO - Livello Technologies (alex@livello.com)

CERTIFICATIONS

- Product management for AI and Data Science: 365 Data Science
- Certified Data Scientist: 365 Data Science
- Build Chat Applications with OpenAI and LangChain: 365 Data Science
- AWS Machine Learning Speciality: Sundog Education
- GitLab – CI/CD Pipelines & DevOps: Udemy
- Data Science Math Skills: Duke University
- Time Series Analysis, Forecasting, and Machine Learning: Udemy
- Vector Databases with Pinecone: 365 Data Science
- GenerativeAI path: Google
- Fundamentals of Docker and Kubernetes: Scaler